Dimension	Category	Description
		TASK
Objective	Writing	Tasks whose primary objective is to produce textual artifacts. *Examples:* Story (Chakrabarty et al., 2024a; Atmakuru et al., 2024; Lu et al., 2025b), poetry (West and Potts, 2025), essay (Kim et al., 2025a), and head-to-head comparison against a professional fiction author (Marco et al., 2024).
	Problem Solving	Tasks whose primary objective is to reach a correct or valid solution. *Examples:* Math (Ye et al., 2025; Han et al., 2025; Chen et al., 2025), physical / multimodal (Tian et al., 2025; Bhavya et al., 2024).
	Ideation	Tasks whose primary objective is to propose ideas or generate candidate solutions or associations. *Examples:* Brainstorming (Lv et al., 2024; Ruan et al., 2025), association tasks (Chen and Ding, 2023; Qiu and Hu, 2025), TTCT-style divergent-thinking suites (Yao et al., 2024; Sun et al., 2025), and analogy generation (Czinczoll et al., 2022).
Domain	Commonsense	Tasks grounded in everyday knowledge and informal reasoning. *Examples:* Commonsense QA (Talmor et al., 2019), Physical Commonsense Reasoning (Bisk et al., 2019), PlanBench (Valmeekam et al., 2023), RiddleSense (Lin et al., 2021).
	Language	Tasks centered on natural language generation or understanding. *Examples:* SQuAD (Rajpurkar et al., 2016), GLUE (Wang et al., 2019), WritingBench (Wu et al., 2025), Creative Writing v3 (Paech, 2025a).
	Math	Tasks involving mathematical reasoning, QA, computation, or problem solving. *Examples:* GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021a), AIME (Balunović et al., 2026), DeepMath-Creative (Chen et al., 2025).
	Computer (CS)	Tasks requiring computer science knowledge or programming skills. *Examples:* HumanEval (Chen et al., 2021), SWE-Bench (Jimenez et al., 2024).
	Science	Tasks requiring scientific explanation, hypothesis generation, or reasoning. *Examples:* GPQA (Rein et al., 2023), AstaBench (Bragg et al., 2025), scientific-analogy probes (Czinczoll et al., 2022).
	Synthetic	Tasks defined in non-natural-language or symbolic environments. *Examples:* Symbolic language or knowledge tasks (Wang et al., 2024; Nagarajan et al., 2025; Xu et al., 2026b).
	Others	Tasks not covered by the above domains. *Examples:* Game strategy (West and Potts, 2025; Qian et al., 2025), simulation / agentic environments (Jia et al., 2024), or design ideation tasks.
Creativity Elicitation Operation	Prescription	Creativity is demanded through the task objective or explicit instruction. *Examples:* Creative story writing (Chakrabarty et al., 2024a; Atmakuru et al., 2024; Lu et al., 2025b; West and Potts, 2025), brainstorming (Ruan et al., 2025), prompts requiring novel solutions (Rabeyah et al., 2024; Ye et al., 2025), alternative tool use (Rabeyah et al., 2024).
	Curation	Creativity is induced by posing inputs that are themselves rare or unusual, such that standard heuristics are insufficient to produce a valid output and elicit creativity to finish. *Examples:* Brainteasers (Han et al., 2025), riddles (Lin et al., 2021), creative writing premise, puns and wordplays.
	Constraint	Creativity is induced by adding explicit requirements or constraints to the input that eliminate the most typical outputs, eliciting the model to depart from its common output. *Examples:* Constrained story writing (Atmakuru et al., 2024), constrained code generation (Lu et al., 2025c).
Targeted Level	Surface (Form)	The task implicitly or explicitly demands creativity at the lexical or stylistic level. *Examples:* Stylistic rewriting tasks (Gómez-Rodríguez and Williams, 2023; West and Potts, 2025), wordplay.
	Meaning (Content)	The task implicitly or explicitly demands creativity at the level of new semantic combinations or content-level variation. *Examples:* Story writing (Gómez-Rodríguez and Williams, 2023; Chakrabarty et al., 2024a; Atmakuru et al., 2024; Lu et al., 2025b; West and Potts, 2025), divergent association tasks (Chen and Ding, 2023; Qiu and Hu, 2025), TTCT-style ideation (Yao et al., 2024).
	Strategy (Concept)	The task implicitly or explicitly demands creativity at the level of novel ideas, concepts, or solution strategies. *Examples:* Creative problem-solving or algorithm design tasks (Tian et al., 2025; Ye et al., 2025; Han et al., 2025), creative simulation / agentic tasks (Jia et al., 2024).
		METRIC
Reference Source	Self Samples	Evaluation relative to the model's own generated samples at test time. *Examples:* Intra-model diversity (Jiang et al., 2025; Zhang et al., 2025a; Atmakuru et al., 2024).
	Training Corpus	Evaluation relative to (a proxy of) the model's training distribution. *Examples:* Web-scale corpus comparison via n-gram or infinigram lookup (Lu et al., 2025b; Padmakumar et al., 2025; Saakyan et al., 2026).
	Human Single	Evaluation from an individual human's judgment relative to their own knowledge. *Examples:* User rating (Stiennon et al., 2020; Ouyang et al., 2022).
	LLM Single	Evaluation from another LLM relative to its output or judgment on its own knowledge. *Examples:* LLM-as-a-judge, inter-model diversity metrics (Jiang et al., 2025).
	Human Community	Evaluation from a population of humans relative to their own knowledge. *Examples:* Crowd-sourced human evaluation (Chhun et al., 2022; Karpinska et al., 2021); story evaluation by multiple literary experts (Marco et al., 2024); group-level TTCT comparisons against human populations (Sun et al., 2025).
	LLM Community	Evaluation from a population of other LLMs relative to their output or judgment on their own knowledge. *Examples:* Multi-model ensembling and multi-agent evaluation (Verga et al., 2024; Chan et al., 2024); multi-LLM critic ensembles for creative writing (Bae and Kim, 2024).
	Human Historical	Evaluation relative to historical human artifacts. *Examples:* This is an imaginary setting; approximations include internet-scale or library-scale databases.
Output	Process	Evaluation considers intermediate steps, plans, drafts, or actions produced during task execution, rather than or in addition to the final output. *Examples:* Reasoning trace (Han et al., 2025), iterative drafting process (Atmakuru et al., 2024), separation of direction- vs. process-accuracy for mathematical reasoning (Chen et al., 2025).
	Outcome	Evaluation considers only the final output of the model. *Examples:* Final answer (Lu et al., 2025b; Tian et al., 2025; Zhang et al., 2025a), final story.
Type	Monolithic	Creativity is treated as a unitary construct and measured directly. *Examples:* Direct human or LLM creativity ratings (Gómez-Rodríguez and Williams, 2023; Kim et al., 2025a; Cao et al., 2026).
	Compositional	Creativity is defined through explicit theoretical components (e.g., novelty and value), each operationalized independently. *Note*: We only discuss the novelty component in this research and consider the value component out of our scope. In experiments, we apply the most widely used metrics for the value component (e.g., reward for writing, accuracy for reasoning). *Examples:* Novelty $\times$ quality decompositions (Zhang et al., 2025a; Padmakumar et al., 2025); TTCT four-factor decomposition (Fluency / Flexibility / Originality / Elaboration) (Yao et al., 2024); convergent $\times$ divergent decomposition for code (Lu et al., 2025c).
Granularity	Token	Evaluation at the level of individual tokens. *Examples:* Token probability or entropy (Holtzman et al., 2020); token-level attention-window scoring for creativity detection (Kim et al., 2025b).
	N-gram	Evaluation over contiguous token sequences. *Examples:* Distinct-n (Li et al., 2016); n-gram corpus-novelty / Creativity Index (Lu et al., 2025b); infinigram-based novelty (Saakyan et al., 2026).
	Sentence	Evaluation at the sentence level. *Examples:* Sentence embedding similarity (Reimers and Gurevych, 2019; Tevet and Berant, 2021).
	Step	Evaluation at the level of procedural or reasoning steps, often composed of multiple sentences. *Examples:* Step-wise entailment evaluation (Prasad et al., 2023; Golovneva et al., 2023).
	Paragraph	Evaluation over the paragraph level, which often is the entire generation. *Examples:* Story evaluation (Chakrabarty et al., 2024a; Atmakuru et al., 2024).
Note: Granularity defines the structural unit at which variation is assessed. Well-designed evaluations align this choice with the task's Targeted Level (e.g., token- or n-gram-level metrics for surface creativity, paragraph-level metrics for content or strategy creativity). Misalignment between granularity and targeted level sometimes appears in existing work which may contribute to inconsistent findings across benchmarks.		
Judge	Deterministic	Rule-based procedures without trainable parameters. *Examples:* BLEU, ROUGE, distinct-n (Papineni et al., 2002; Lin, 2004; Li et al., 2016).
	Learned	Trained models producing evaluative signals. *Examples:* Reward models (Ouyang et al., 2022); trained embedding-based scorers (Zhang et al., 2020); creativity-specific trained scoring models such as R2C and CLAWS (Kim et al., 2025a,b); fine-tuned creativity evaluators (Cao et al., 2026).
	LLM-as-a-Judge	Evaluation performed by prompting an LLM under a judging protocol. *Examples:* Rubric-based LLM judging (Zheng et al., 2023; Liu et al., 2023); multi-critic LLM ensembles for creative writing (Bae and Kim, 2024).
	Human	Evaluation performed by human annotators. *Examples:* Expert evaluation (Chakrabarty et al., 2024a) or crowd-sourced evaluation (Chhun et al., 2022).